

Module 18

IOT and OT Hacking

What is IoT (Internet of Things)?

IoT (Internet of Things) refers to smart devices connected to the internet that can collect and exchange data. These include:

- Smart home devices: Alexa, smart TVs, lights, thermostats
- Wearables: Smartwatches, fitness trackers
- Industrial IoT: Smart meters, connected vehicles, medical devices

 **Purpose: Automation, data collection, remote control, real-time monitoring**

What is OT (Operational Technology)?

OT (Operational Technology) refers to hardware and software that controls **industrial systems and infrastructure**. These are often used in:

- Power plants
- Manufacturing (SCADA systems, PLCs)
- Oil & gas industries
- Water treatment facilities

 **Purpose: Control physical devices and processes (e.g., open a valve, turn on a motor)**

Hacking IoT & OT: Explained

IoT Hacking

Since IoT devices are connected to the internet and often poorly secured, they are attractive targets.

Common IoT Vulnerabilities:

- Default credentials (admin/admin)
- Insecure APIs
- Unpatched firmware
- Weak encryption

Example IoT Attacks:

- **Mirai Botnet:** Compromised thousands of IoT devices to launch DDoS attacks.
- **Smart camera hacks:** Attackers gain access to private video streams.

Tools for IoT Hacking:

- **Shodan** (search engine for IoT devices)
 - **Wireshark** (packet capture)
 - **Burp Suite** (API testing)
 - **Binwalk** (firmware analysis)
-

OT Hacking

OT systems were traditionally isolated (air-gapped), but now many are connected to IT networks, increasing the attack surface.

Common OT Vulnerabilities:

- Legacy systems with no security in design

- No authentication or encryption
- Insecure protocols (Modbus, DNP3)
- Poor segmentation between IT and OT networks

Example OT Attacks:

- **Stuxnet (2010)**: Targeted Iranian nuclear centrifuges by manipulating PLCs.
- **Triton (2017)**: Attacked safety systems in a petrochemical plant.
- **Ukraine power grid attack (2015)**: Shut down power for 230,000+ people.

Tools for OT Hacking:

- **PLCScan**: Scan for programmable logic controllers
- **ModScan/Modbus-cli**: Attack Modbus protocols
- **Wireshark**: Monitor ICS traffic
- **Metasploit**: Exploit known OT system vulnerabilities

Key Differences Between IoT and OT Hacking

Feature	IoT Hacking	OT Hacking
Target	Smart devices	Industrial control systems
Motivation	Data theft, botnets, surveillance	Sabotage, disruption, espionage
Protocols	HTTP, MQTT, CoAP	Modbus, DNP3, OPC, SCADA-specific
Attack Surface	Consumer-grade, internet-connected	Industrial-grade, often legacy systems
Risk Level	Privacy breach, device control	Physical damage, safety risks

Securing IoT & OT:

- Change default passwords
 - Network segmentation (IT/OT separation)
 - Patch and update firmware/software
 - Use firewalls and IDS (Intrusion Detection Systems)
 - Monitor traffic and behavior anomalies
-

❖ Shodan uses for IOT and OT hacking

Shodan : Shodan is often called the "search engine for hackers" because it allows users to discover internet-connected devices — especially **IoT (Internet of Things)** and **OT (Operational Technology)** systems — across the globe. Here's how Shodan is used in **IoT and OT hacking (both offensive and defensive purposes)**:

🔍 Uses of Shodan in IoT and OT Hacking

✅ 1. Device Discovery

- **What it does**: Finds internet-exposed IoT/OT devices like CCTV cameras, routers, SCADA systems, industrial control systems (ICS), smart TVs, printers, HVAC, etc.
- **Use case**: Identify vulnerable devices for penetration testing or red teaming.

✅ 2. Service Enumeration

- **What it does**: Lists open ports and services running (e.g., HTTP, Telnet, FTP, Modbus, BACnet).
- **Use case**: Helps attackers or security researchers identify potentially weak services like unauthenticated access or outdated software.

✓ 3. Vulnerability Assessment

- **What it does:** Shodan tags known vulnerabilities (CVEs) linked to specific devices and firmware versions.
- **Use case:** Find systems vulnerable to CVEs like CVE-2020-9054 (routers), or CVE-2015-1013 (SCADA systems).

✓ 4. Geo-Location & Target Profiling

- **What it does:** Provides IP, country, organization, and sometimes GPS location.
- **Use case:** Identify critical infrastructure by region (e.g., water plants, power grids in a specific country).

✓ 5. Industrial Protocol Exposure

- **What it does:** Detects ICS protocols like:
 - **Modbus**
 - **DNP3**
 - **Siemens S7**
 - **BACnet**
 - **HART**
- **Use case:** OT security testing; some of these protocols lack authentication and can be misused if exposed online.

✓ 6. Banner Grabbing

- **What it does:** Shodan shows banners from services (e.g., software version, device type).
- **Use case:** Used for fingerprinting devices and finding misconfigured or unpatched systems.

✓ 7. Automation in Red Teaming

- **What it does:** Can be integrated with tools like Shodan API, Nmap, or Metasploit for automated reconnaissance.
- **Use case:** Speeds up target gathering in offensive security operations.

Use in Defensive Security (Blue Team)

- **Asset Management:** Detect unauthorized or rogue IoT/OT devices connected to the internet.
 - **Exposure Monitoring:** Continuously monitor your organization's public IP range.
 - **Threat Hunting:** Look for signs of exposed ICS/SCADA systems that should not be public.
 - **Compliance Checks:** Ensure devices are not leaking sensitive information or control interfaces.
-

Example Queries

- port:502 product:modbus → Find Modbus systems.
 - country:"IN" port:1911 → BACnet systems in India.
 - port:23 org:"XYZ Company" → Open Telnet in a specific org.
 - product:"WebcamXP" → Exposed webcams.
-

➤ Here are some example and screenshot of using shodan for IOT and OT devices.

Example 1: Search for Webcams

Search Query:

```
product:"webcamXP"
```

Explanation:

- Finds cameras running **webcamXP**, often with a web interface and sometimes **no password protection**.
- You may see preview images, IPs, and open ports.

product:"webcamXP" - Shodan

Shodan in IoT OT Hacking

shodan.io/search?query=product%3A"webcamXP"

SHODAN

Explore

Downloads

Pricing

product:"webcamXP"

Account

TOTAL RESULTS

132

TOP COUNTRIES



United States	34
Germany	14
Russian Federation	14
Spain	7
France	5
More...	

TOP PORTS

8080	46
8090	12
80	10
81	4

View Report

Browse Images

View on Map

Advanced Search

Access Granted: Want to get more out of your existing Shodan account? Check out everything you have access to.

webcamXP 5

24.126.96.142

Comcast Cable Communications Holdings, Inc.

United States, Milford

IoT

HTTP/1.1 200 OK

Connection: close

Content-Type: text/html; charset=utf-8

Content-Length: 7327

Cache-control: no-cache, must revalidate

Date: Mon, 23 Jun 2025 12:50:20 GMT

Expires: Mon, 23 Jun 2025 12:50:20 GMT

Pragma: no-cache

Server: webcamXP 5



2025-06-23T12:50:21.866511

26°C

Partly sunny

Search

ENG IN

18:54

23-06-2025

product:"webcamXP" - Shodan

Shodan in IoT OT Hacking

shodan.io/search?query=product%3A"webcamXP"

SHODAN

Explore

Downloads

Pricing

product:"webcamXP"

Account

TOP OPERATING SYSTEMS

Windows

11

Ubuntu

1

webcamXP 5

184.57.102.6

syn-184-057-102-006.res.spectrum.com

Charter Communications Inc.

United States, Erie

IoT

HTTP/1.1 200 OK

Connection: close

Content-Type: text/html; charset=utf-8

Content-Length: 7327

Cache-control: no-cache, must revalidate

Date: Sat, 05 Jan 2002 15:54:22 GMT

Expires: Sat, 05 Jan 2002 15:54:22 GMT

Pragma: no-cache

Server: webcamXP 5



2025-06-23T09:28:46.365444

26°C

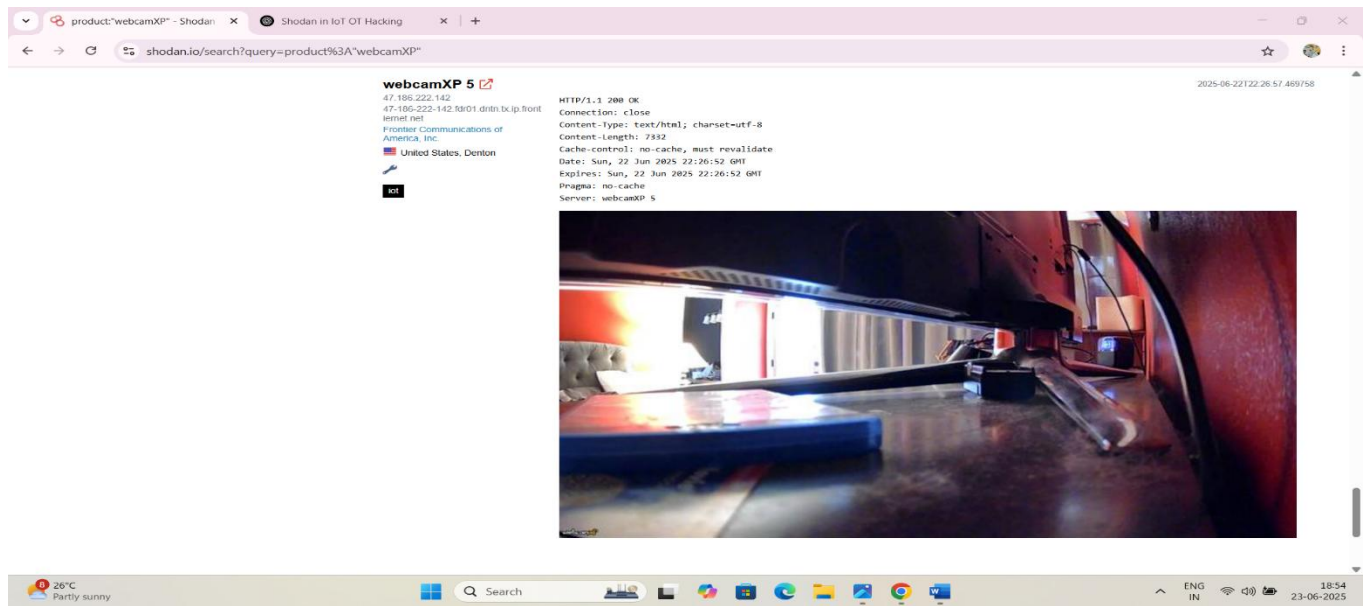
Partly sunny

Search

ENG IN

18:54

23-06-2025



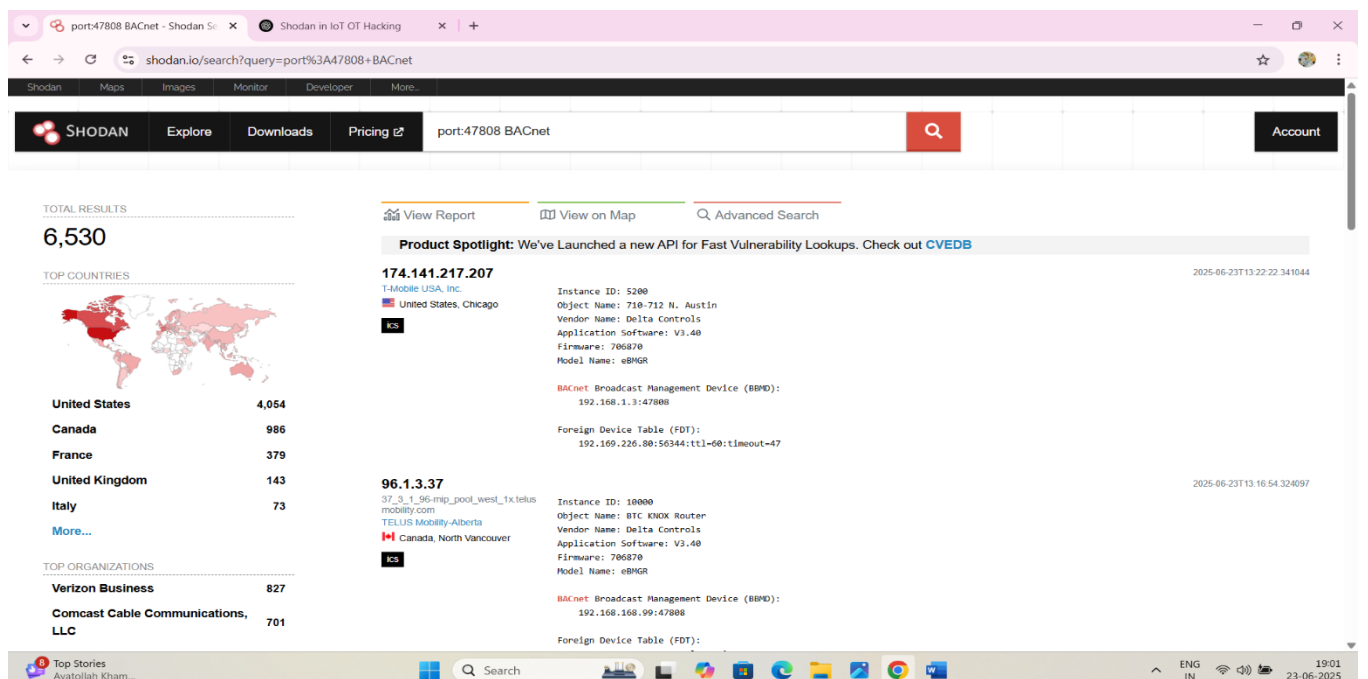
🔍 Example 2: Exposed BACnet Devices

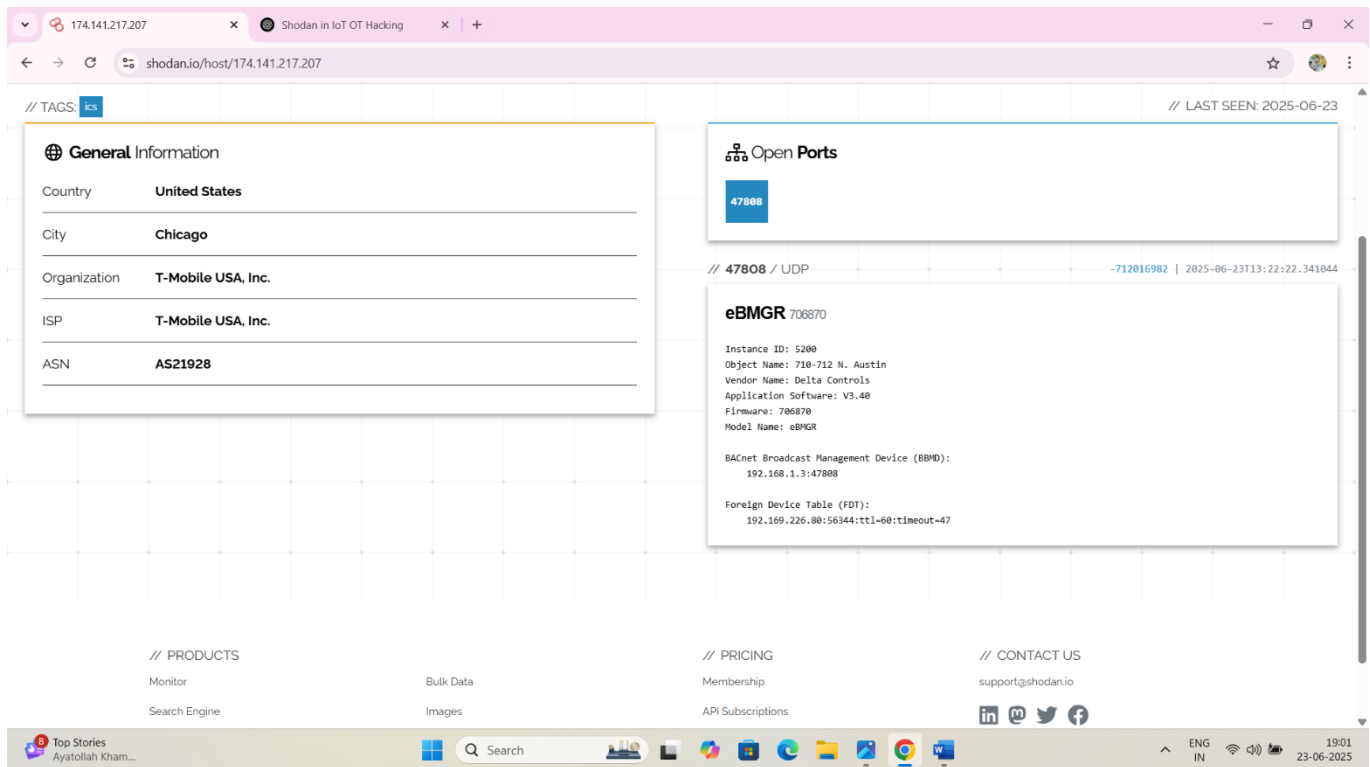
Search Query:

port:47808 BACnet

Explanation:

- BACnet is used in building automation systems (HVAC, lighting).
- Exposed BACnet devices can be misused for physical disruption.





The screenshot shows a web browser window with the URL `shodan.io/host/174.141.217.207`. The page displays information for the IP address 174.141.217.207. The main content area is divided into two columns. The left column, titled "General Information", contains a table with the following data:

General Information	
Country	United States
City	Chicago
Organization	T-Mobile USA, Inc.
ISP	T-Mobile USA, Inc.
ASN	AS21928

The right column, titled "Open Ports", shows a list of open ports. The first port listed is 47808 / UDP. Below this, there is a section for "eBMGR 706870" with the following details:

- Instance ID: 5200
- Object Name: 710-712 N. Austin
- Vendor Name: Delta Controls
- Application Software: V3.40
- Firmware: 706870
- Model Name: eBMGR

Below this, there is a section for "BACnet Broadcast Management Device (BBMD)" with the following details:

- 192.168.1.3:47808

Below this, there is a section for "Foreign Device Table (FDT)" with the following details:

- 192.169.226.80:56344:ttl=60:timeout=47

The bottom of the page features a navigation bar with links for "PRODUCTS", "PRICING", and "CONTACT US". The "PRODUCTS" section includes links for "Monitor", "Bulk Data", and "Search Engine". The "PRICING" section includes links for "Membership" and "API Subscriptions". The "CONTACT US" section includes a link for "support@shodan.io".

🔍 Example 3: Find Devices in India Running Telnet

Search Query:

`port:23 country:"IN"`

Explanation:

- Lists devices in India running Telnet — often routers or IoT devices with default credentials.

port23 country:"IN" - Shodan S

Shodan in IoT OT Hacking

shodan.io/search?query=port%3A23+country%3A"IN"

TOTAL RESULTS

37,633

View Report

View on Map

Advanced Search

Access Granted: Want to get more out of your existing Shodan account? Check out everything you have access to.

182.78.21.41

BHARTI-AIRTEL

India, Gurugram

Router login:

2025-06-23T13:32:46.249465

103.38.182.190

SAITECH SERVICES

India, Padampur

Login:

2025-06-23T13:25:15.314213

203.192.197.220

dhcp-192-197-220.in2cable.com

Indusind Media And Communication Ltd

192.0.0.64 login:

2025-06-23T13:25:14.304523

14.195.192.192

static-192-195-14-tataidc.co.in

Tata Teleservices Limited -GSM Division

India, Chennai

Firewall requires that you agree to the disclaimer using a web browser before proceeding with this service\r\n

2025-06-23T13:24:59.688012

182.72.91.57

nsg-static-057.91.72-182.airtel.in

ETC MANAGEMENT SERVICES I

India, Bengaluru

BYSYN-BLL101-IN-RTR1001 login:

2025-06-23T13:23:06.428080

182.79.31.143

DALMA BHARAT ENTERPRISES

India, Chennai

THIS SYSTEM IS FOR THE USE OF AUTHORIZED USERS ONLY

USERS HAVE NO EXPLICIT OR IMPLICIT EXPECTATIONS OF PRIVACY

2025-06-23T13:22:49.442768

TOP CITIES

Mumbai

5,281

Delhi

4,244

Bengaluru

2,629

Gurugram

2,267

Kolkata

2,189

More...

TOP ORGANIZATIONS

BHARTI-AIRTEL

3,984

Bharti Televentures Ltd. - ABTS

1,828

Broadband Multiplay Project, O/o DGM BB, NOC BSNL Bangalore

1,480

BHARTI TELENET LTD. NEW DELHI

1,280

Reliance Jio Infocomm Limited

1,171

More...

TOP PRODUCTS

Busybox telnetd

5,084

Cisco router telnetd

2,361

MPR-L8 3G mobile router telnetd

758

Light rain

At night

Search

ENG IN

19:06

23-06-2025

Example 4: Search for Open Industrial Devices (Modbus)

shodan search port:502 modbus

port:502 modbus - Shodan S

Shodan in IoT OT Hacking

shodan.io/search?query=+port%3A502+modbus

Shodan Maps Images Monitor Developer More...

SHODAN Explore Downloads Pricing port:502 modbus Account

TOTAL RESULTS

87

View Report

View on Map

Advanced Search

Product Spotlight: Free, Fast IP Lookups for Open Ports and Vulnerabilities using InternetDB

73.178.148.76

c-73-178-148-76.hsd1.nj.comcast.net

Comcast IP Services, L.L.C.

United States, Linwood

Unit ID: 0

Unit ID: 1

Unit ID: 255

-- Device Identification: Modbus/TCP to RTU Bridge Bridge V3.3.25.0RC6

2025-06-23T07:34:56.298358

83.187.119.143

m83-187-119-143.cust.tele2.lt

Mobile Services Lithuania

Lithuania, Vilnius

Unit ID: 1

-- Slave ID Data: 7* 7ME692 - Siemens MAG6000 --- MODBUS RTU (372aff18374545363932002d0001ff005369656d656e73000000000004

2025-06-23T06:54:49.015324

101.201.66.236

Allyun Computing Co., LTD

China, Beijing

Unit ID: 0

-- Slave ID Data: (03000000)

-- Device Identification: Witte Software Modbus Slave V9.4.3, Build 2293 https://www.modbustools.com/

Unit ID: 1

-- Slave ID Data: (03000000)

-- Device Identification: Witte Software Modbus Slave V9.4.3, Build 2293 https://www.modbustools.com/

...

2025-06-23T06:26:40.273407

TOP COUNTRIES

India

44

United States

22

Australia

5

Romania

5

France

2

More...

TOP ORGANIZATIONS

Internet Service Provider

7

Verizon Business

7

RailTel Delhi

5

Sports headline

Oklahoma City T...

Search

ENG IN

19:11

23-06-2025

🔧 Example: Find Cisco Devices

shodan search "cisco" country:"IN"

Shodan

Maps

Images

Monitor

Developer

More...

SHODAN

Explore

Downloads

Pricing

"cisco" country:"IN"

Q

Account

TOTAL RESULTS

59,711

TOP CITIES

Mumbai

13,037

Bengaluru

6,668

Chennai

6,143

Delhi

5,254

Kolkata

3,920

More...

TOP PORTS

161

46,340

22

6,253

443

2,666

80

1,561

23

732

More...

TOP ORGANIZATIONS

Reliance Jio Infocomm Limited

3,253

Sify Limited

2,847

BHARTI TELENET LTD. NEW DELHI

2,652

More...

View Report

View on Map

Advanced Search

Product Spotlight: We've Launched a new API for Fast Vulnerability Lookups. Check out [CVEDB](#)

14.143.42.81

14.143.42.81 static-mumbai.vsnl.net.
in
Internet Service Provider
India, Mumbai

SNMP:
Versions:
3
EngineID Format: mac
Engine Boots: 2
EngineID Data: a0:0f:37:f0:9c:54
Enterprise: 9
Engine Time: 816 days, 20:05:39
Enterprise Name: ciscoSystems

2025-06-23T13:40:24.08671

125.20.228.203

Bharti Televentures Limited A/c
ABTS MP
India, Gurugram

SNMP:
Versions:
3
Engine Boots: 6
EngineID Data: 000000003006c6cd31747e0
Enterprise: 9
Engine Time: 195 days, 17:12:27
Enterprise Name: ciscoSystems

2025-06-23T13:38:38.826025

122.186.140.191

nsg-corporate-191.140.186.122.airtel
in
BHARTI TELENET LTD. NEW DELHI

SNMP:
Versions:
3

2025-06-23T13:38:00.910092

Watchlist Ideas

Q Search

ENG IN

19:13 23-06-2025

"cisco" country:"IN" - Shodan S

Shodan in IoT OT Hacking

shodan.io/search?query="+cisco"+country%3A"IN"

Reliance Jio Infocomm Limited

3,253

Sify Limited

2,847

BHARTI TELENET LTD. NEW DELHI

2,652

More...

TOP PRODUCTS

ciscoSystems

46,094

Cisco IOS http config

3,618

Cisco Umbrella

755

Cisco SSH

441

PPTP

360

More...

TOP OPERATING SYSTEMS

IOS

4,078

Cisco IOS XE 16.9.3

338

Cisco IOS 15.6(2)SP4

135

Cisco IOS 15.4(3)M3

76

Cisco IOS 15.5(3)S4b

34

More...

152.52.111.188

BHARTI-AIRTEL
India, Delhi

SNMP:
Versions:
3
Engine Boots: 3
EngineID Data: 000000003004cecf8a0260
Enterprise: 9
Engine Time: 242 days, 17:24:04
Enterprise Name: ciscoSystems

2025-06-23T13:37:56.304545

128.185.40.4

BHARTI-AIRTEL
India, Noida

SNMP:
Versions:
3
Engine Boots: 4
EngineID Data: 000000003006c6cd31c9670
Enterprise: 9
Engine Time: 417 days, 15:54:30
Enterprise Name: ciscoSystems

2025-06-23T13:36:36.273624

125.20.0.100

WORLD PHONE INTERNET SERVICES PVT LTD
India, Noida

SNMP:
Versions:
3
Engine Boots: 4
EngineID Data: 00000000300f4db6efef620
Enterprise: 9
Engine Time: 282 days, 15:34:33
Enterprise Name: ciscoSystems

2025-06-23T13:35:22.453580

122.15.74.175

Vodafone Idea Ltd
India, Manappakkam

SNMP:
Versions:
3

2025-06-23T13:34:24.495665

Watchlist Ideas

Q Search

ENG IN

19:13 23-06-2025



Installation and uses of Can-utils

Can-utils :

can-utils are used to interact with Controller Area Network (CAN) buses, primarily in Linux systems. They provide a set of command-line tools for sending, receiving, recording, and replaying CAN traffic. This is crucial for developing, debugging, and testing applications that rely on CAN communication, particularly in automotive and industrial automation contexts.

Here's a more detailed breakdown:

Key functionalities and uses:

- **Monitoring CAN traffic:**

Tools like `candump` allow you to capture and display CAN messages in real-time, which is essential for understanding communication patterns on a CAN bus.

- **Generating CAN traffic:**

`cansend` and `cangen` allow you to send specific CAN messages or generate a stream of random CAN data, which is useful for testing and simulating CAN devices.

- **Replaying CAN traffic:**

`canplayer` enables you to replay captured CAN logs, which is helpful for simulating scenarios or debugging issues that occurred in the past.

- **Analyzing CAN traffic:**

`cansniffer` provides a way to analyze CAN traffic, often used in conjunction with `candump` to understand the content of the messages.

- **Controlling CAN interfaces:**

`ip` and other tools allow you to configure and manage the CAN interface on your system, including setting the bitrate and enabling/disabling the interface.

- **Testing and debugging:**

can-utils is heavily used for testing and debugging CAN-based systems. By monitoring, simulating, and replaying traffic, developers can identify and resolve communication issues.

- **Automotive hacking (with caution):**

can-utils, in combination with other tools, can be used for research and security analysis of vehicle systems. However, it's crucial to use these tools responsibly and ethically.

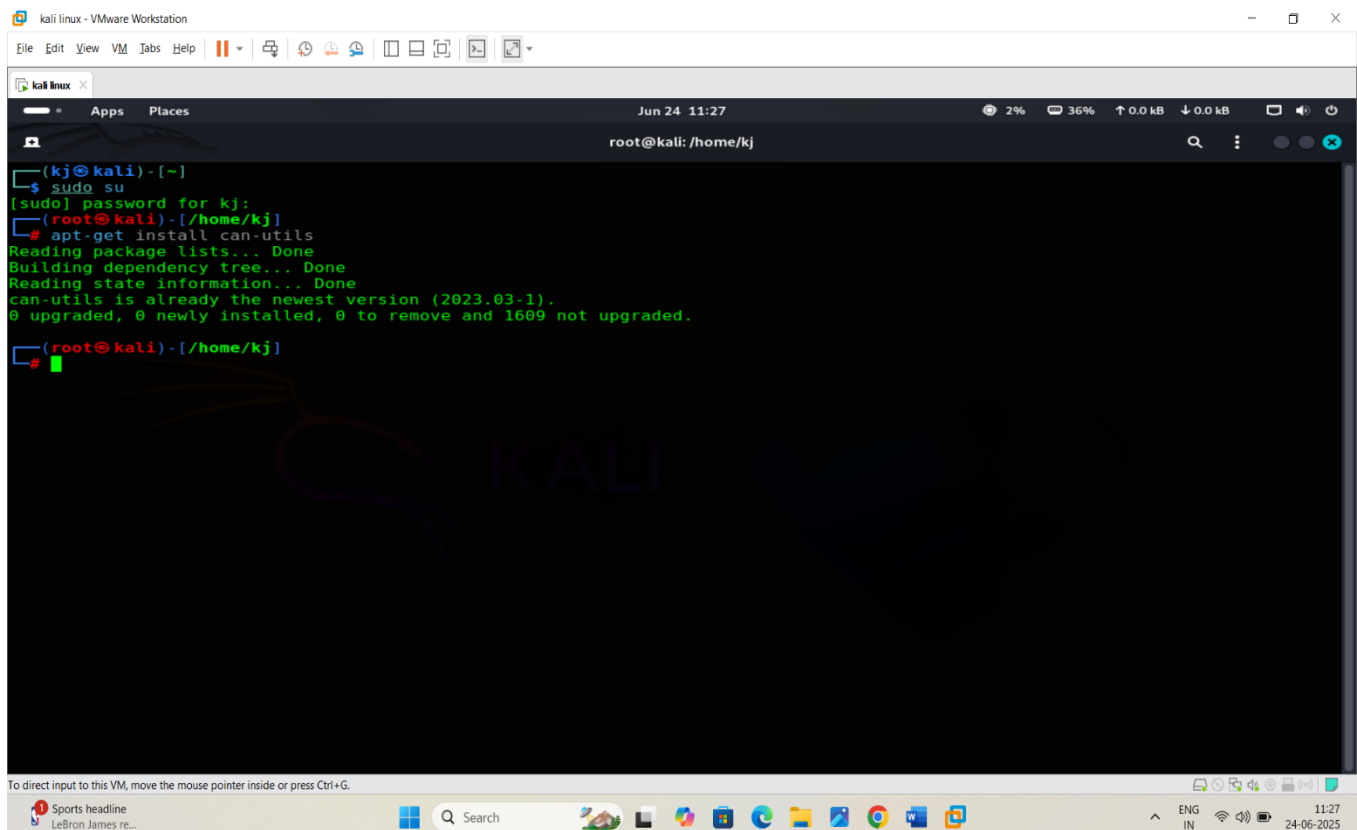
In essence, can-utils acts as a bridge between the Linux operating system and the CAN bus, providing a versatile toolkit for interacting with this crucial network protocol.

Installation :

1. Open kali linux open terminal and type

Apt-get install can-utils

For install the can utils

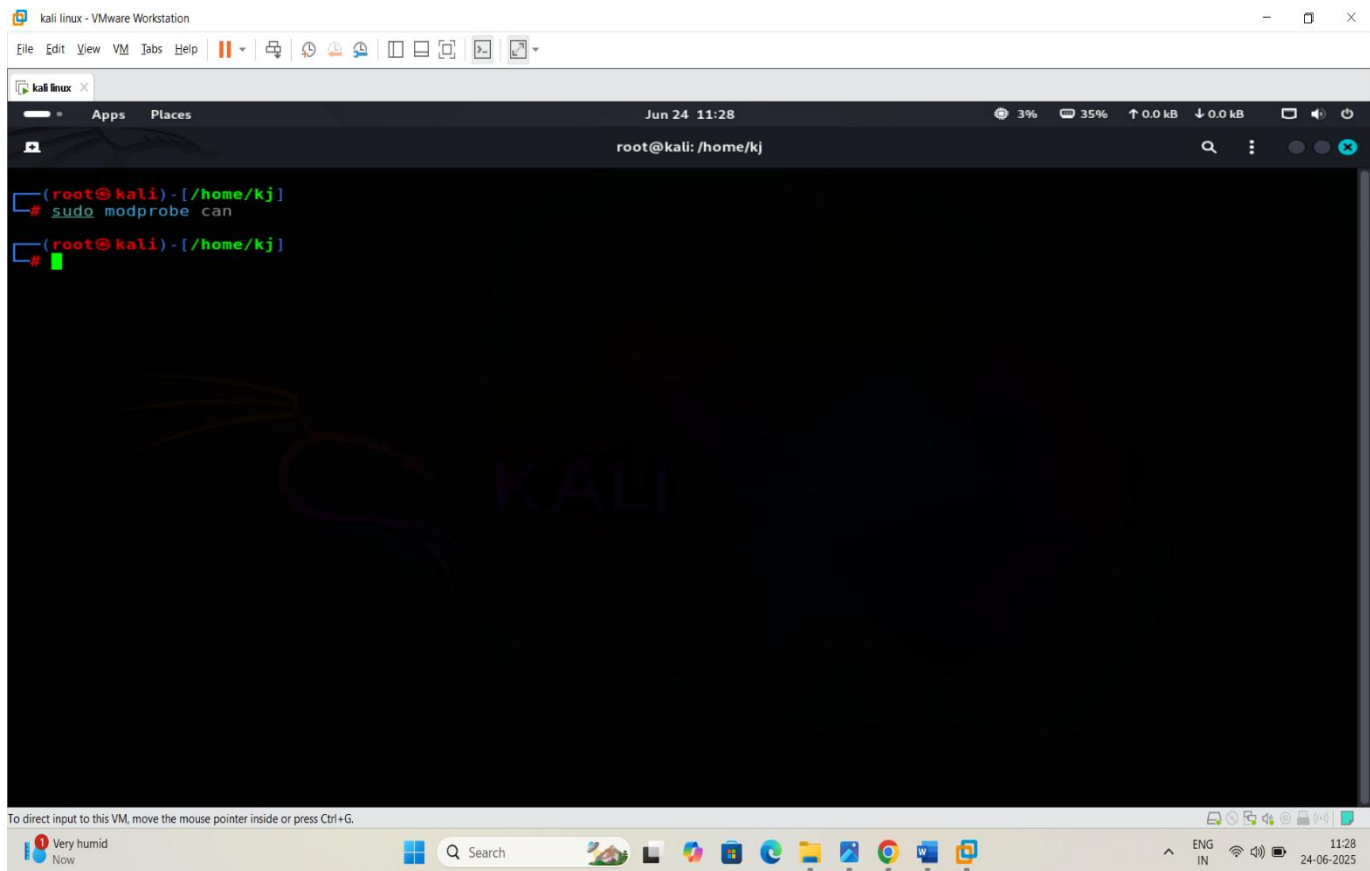


The screenshot shows a Kali Linux terminal window within a VMware Workstation. The terminal output is as follows:

```
(kj@kali) - [~]
$ sudo su
[sudo] password for kj:
(root@kali) - [/home/kj]
# apt-get install can-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
can-utils is already the newest version (2023.03-1).
0 upgraded, 0 newly installed, 0 to remove and 1609 not upgraded.
(root@kali) - [/home/kj]
#
```

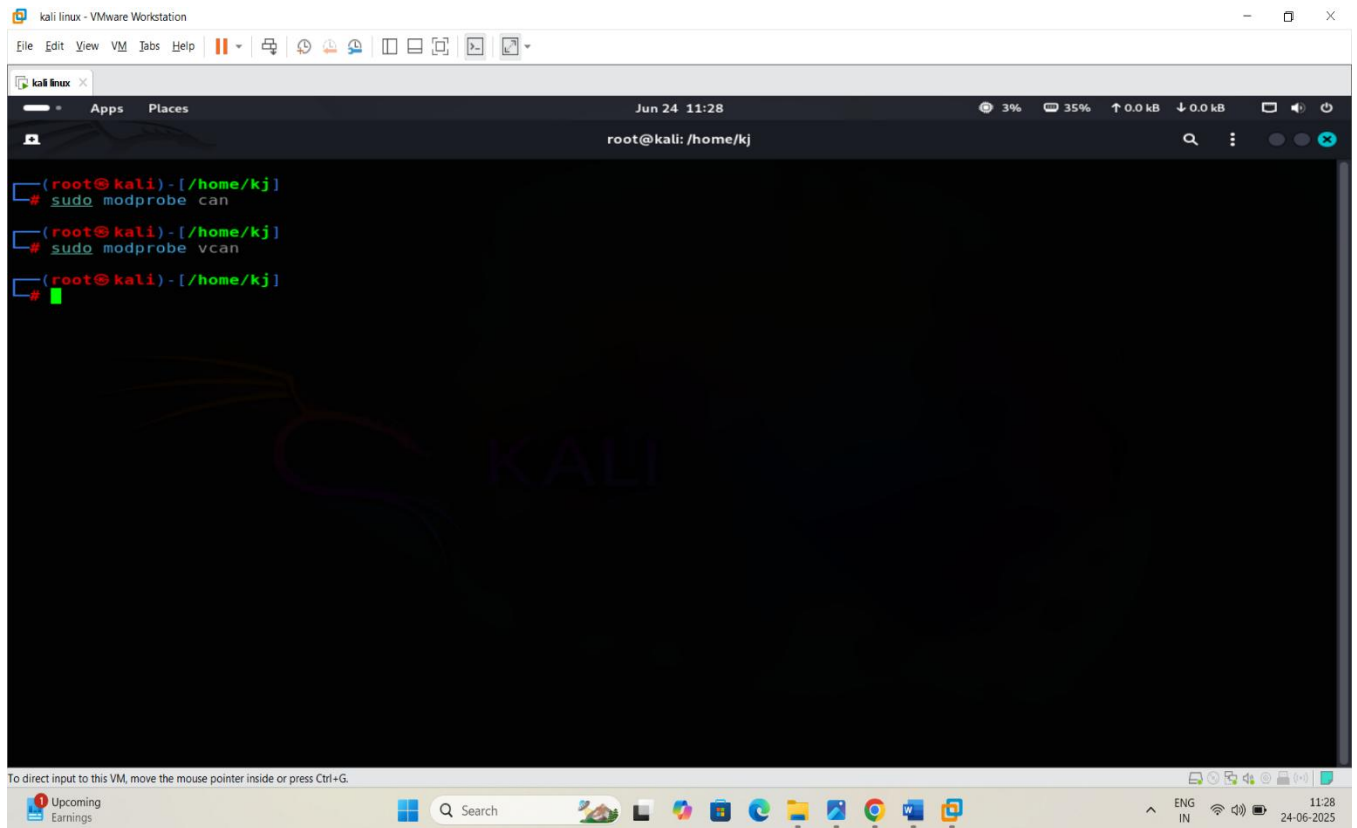
The terminal window has a title bar that reads "kali linux - VMware Workstation". The top of the terminal shows the date and time as "Jun 24 11:27". The bottom of the terminal shows the user "root@kali: /home/kj". The background of the terminal features a large, stylized "KALI" logo.

2. # sudo modprobe can
is used in **CAN (Controller Area Network) communication setup** on Linux systems before using **can-utils**.

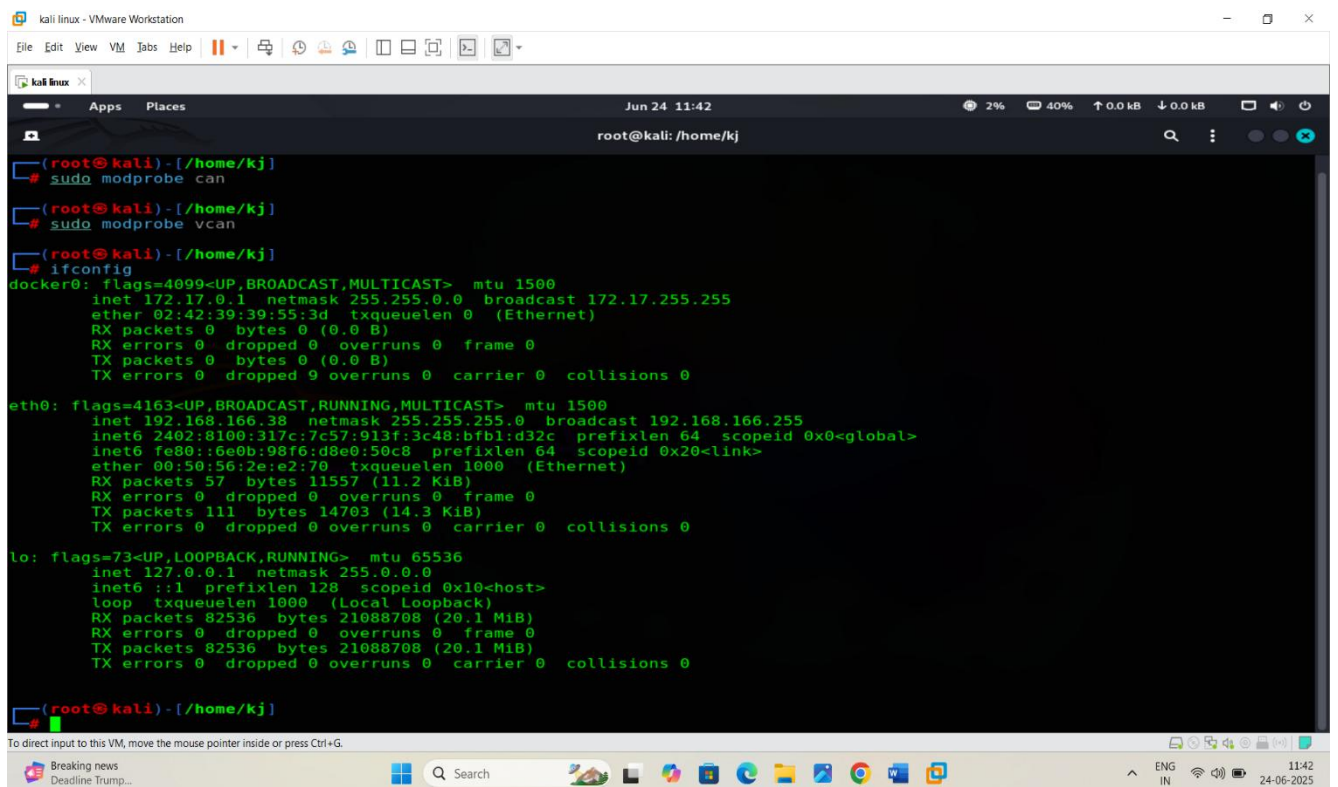


```
kali linux - VMware Workstation
File Edit View VM Tabs Help
kali linux x
Apps Places
Jun 24 11:28
root@kali: /home/kj
(root@kali) - [/home/kj]
# sudo modprobe can
(root@kali) - [/home/kj]
```

3. # Sudo modprobe vcan
It enables the Linux system to support **virtual CAN devices** like **vcan0**. Without loading this module, you **cannot create** a virtual CAN interface



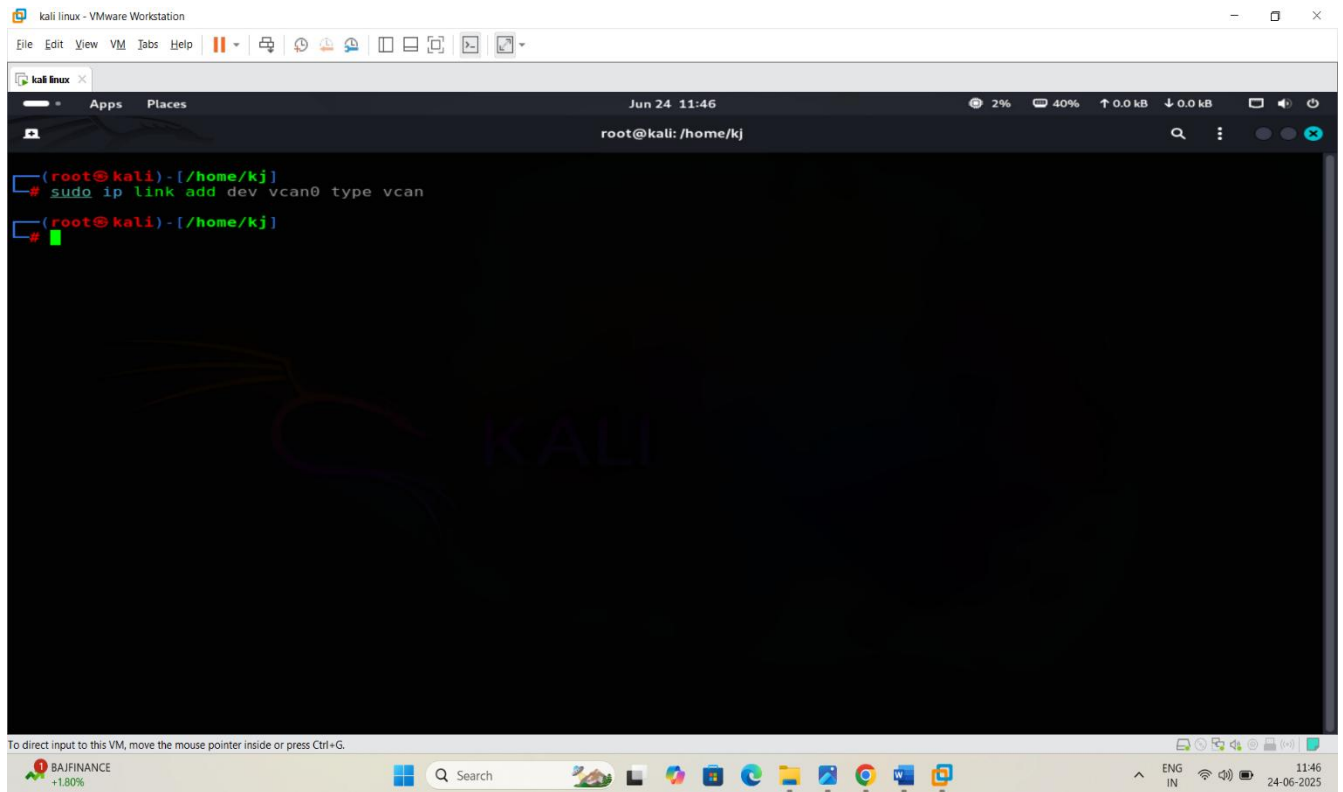
To create new virtual environment:



Here we see no virtual environment is ready

Commands for create new virtual environment

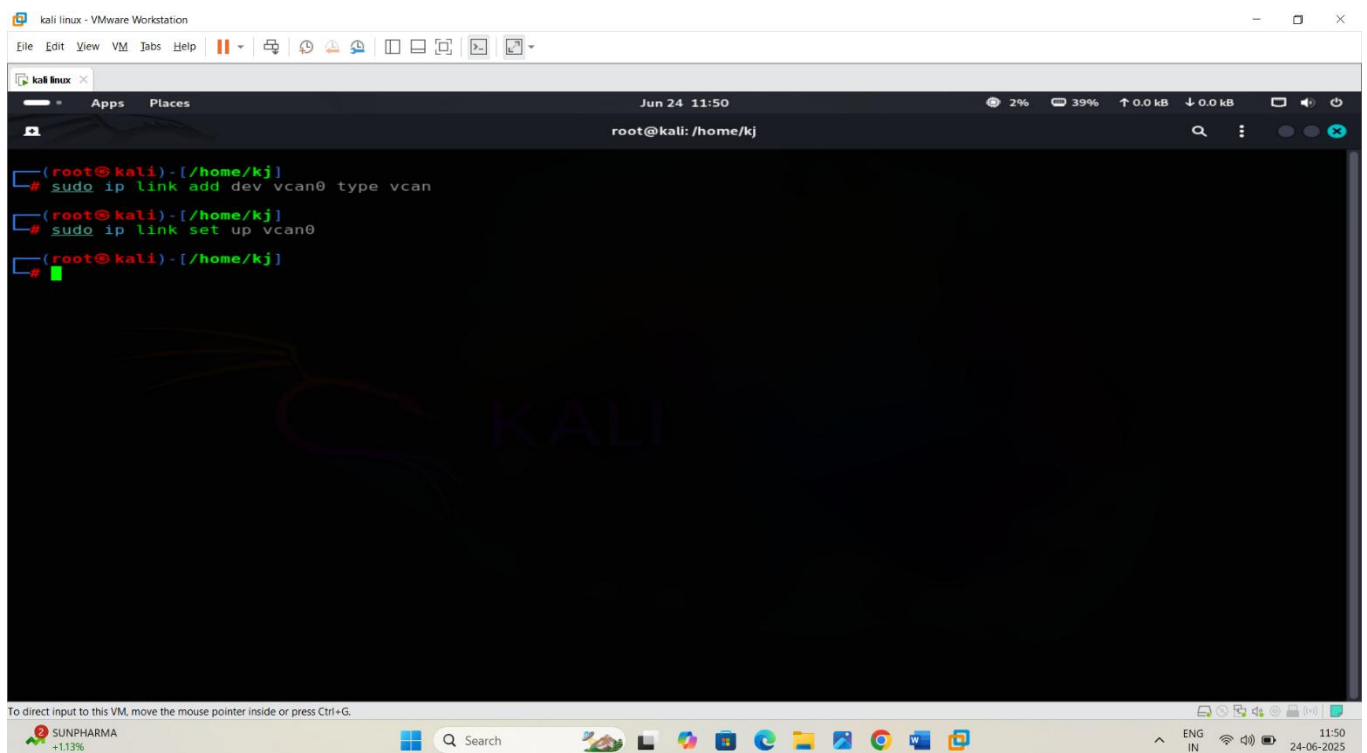
1. # sudo ip link add dev vcan0 type vcan



The screenshot shows a Kali Linux terminal window titled 'kali linux - VMware Workstation'. The terminal prompt is 'root@kali: /home/kj'. The command 'sudo ip link add dev vcan0 type vcan' has been entered and executed. The terminal output shows the command being run successfully. The background of the terminal window features a large, stylized 'KALI' logo.

```
(root@kali)-[/home/kj]
# sudo ip link add dev vcan0 type vcan
(root@kali)-[/home/kj]
#
```

2. # sudo ip link set up vcan0
For bring up the vcan interface

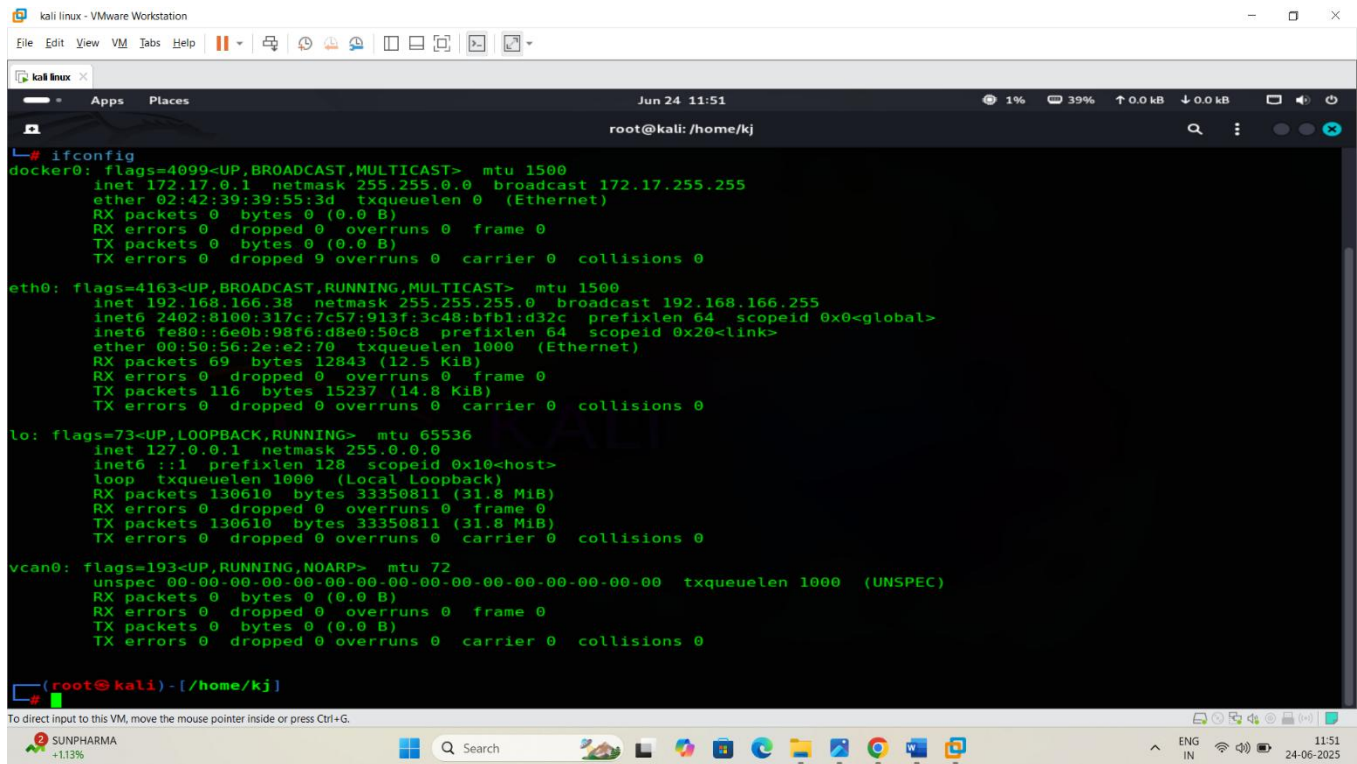


The screenshot shows a Kali Linux terminal window titled 'kali linux - VMware Workstation'. The terminal prompt is 'root@kali: /home/kj'. The command 'sudo ip link set up vcan0' has been entered and executed. The terminal output shows the command being run successfully. The background of the terminal window features a large, stylized 'KALI' logo.

```
(root@kali)-[/home/kj]
# sudo ip link add dev vcan0 type vcan
(root@kali)-[/home/kj]
# sudo ip link set up vcan0
(root@kali)-[/home/kj]
#
```

3. # ifconfig

To verify interface is up



```
root@kali: /home/kj
# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:39:39:55:3d txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 9 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.166.38 netmask 255.255.255.0 broadcast 192.168.166.255
    inet6 2402:8100:317c:7c57:913f:3c48:bfb1:d32c prefixlen 64 scopeid 0x0<global>
    inet6 fe80::6e0b:98f6:d8e0:50c8 prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:2e:e2:70 txqueuelen 1000 (Ethernet)
    RX packets 69 bytes 12843 (12.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 116 bytes 15237 (14.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

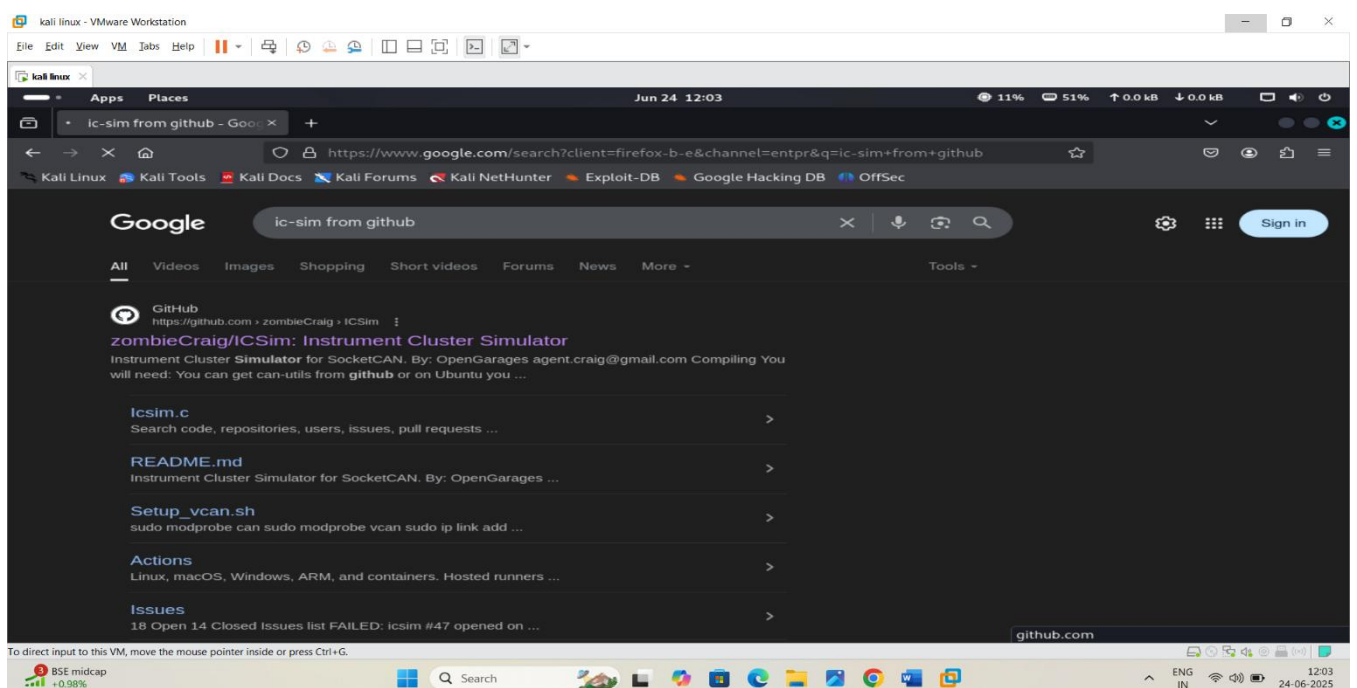
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 130610 bytes 33350811 (31.8 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 130610 bytes 33350811 (31.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

vcan0: flags=193<UP,RUNNING,NOARP> mtu 72
    unspec 00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00 txqueuelen 1000 (UNSPEC)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@kali) - [/home/kj]
```

Here we see the new interface is ready to use .

Now open browser and get IC-Sim from github



IC-Sim : **IC-Sim** (Integrated Circuit Simulator) is an **open-source digital logic circuit simulator**, primarily used for learning and designing logic circuits like gates, flip-flops, multiplexers, counters, etc. It's a lightweight tool designed for **education and hobbyist-level digital circuit simulation**.

Key Features of IC-Sim:

- **Graphical Interface:** You can design logic circuits using a simple GUI.
- **Logic Simulation:** It simulates how digital circuits behave (e.g., AND, OR, NOT, NAND gates).
- **Interactive:** You can toggle inputs and see outputs change in real time.
- **Supports Flip-Flops, Clocks:** Includes sequential circuit elements.
- **Lightweight and Fast:** Great for running on low-end systems or for quick testing

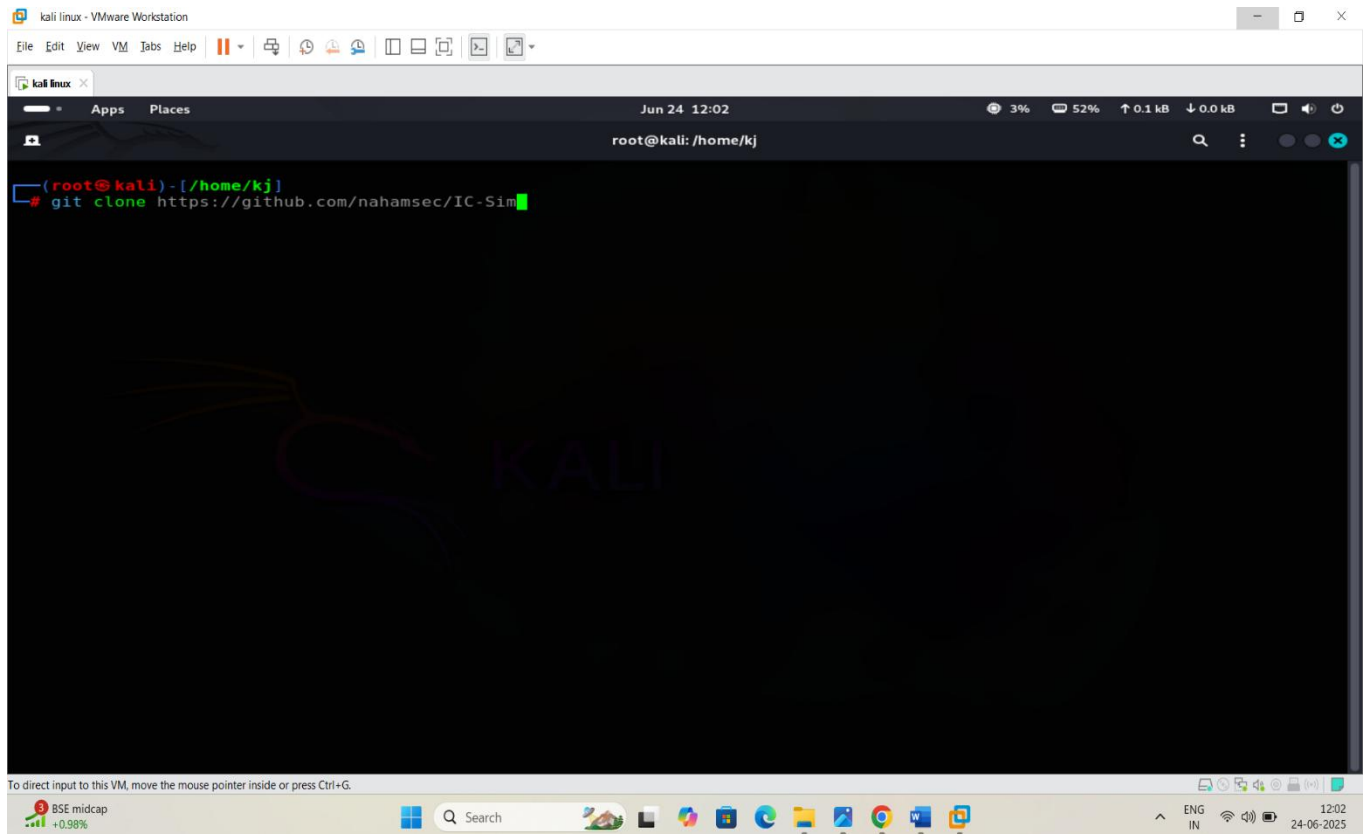
Why Use IC-Sim?

- **Educational Purposes:** It helps in understanding digital electronics and how different logic components work.
- **Beginner-Friendly:** No need for deep HDL knowledge (like Verilog or VHDL).
- **Free & Open Source:** You can modify or enhance the code if needed.

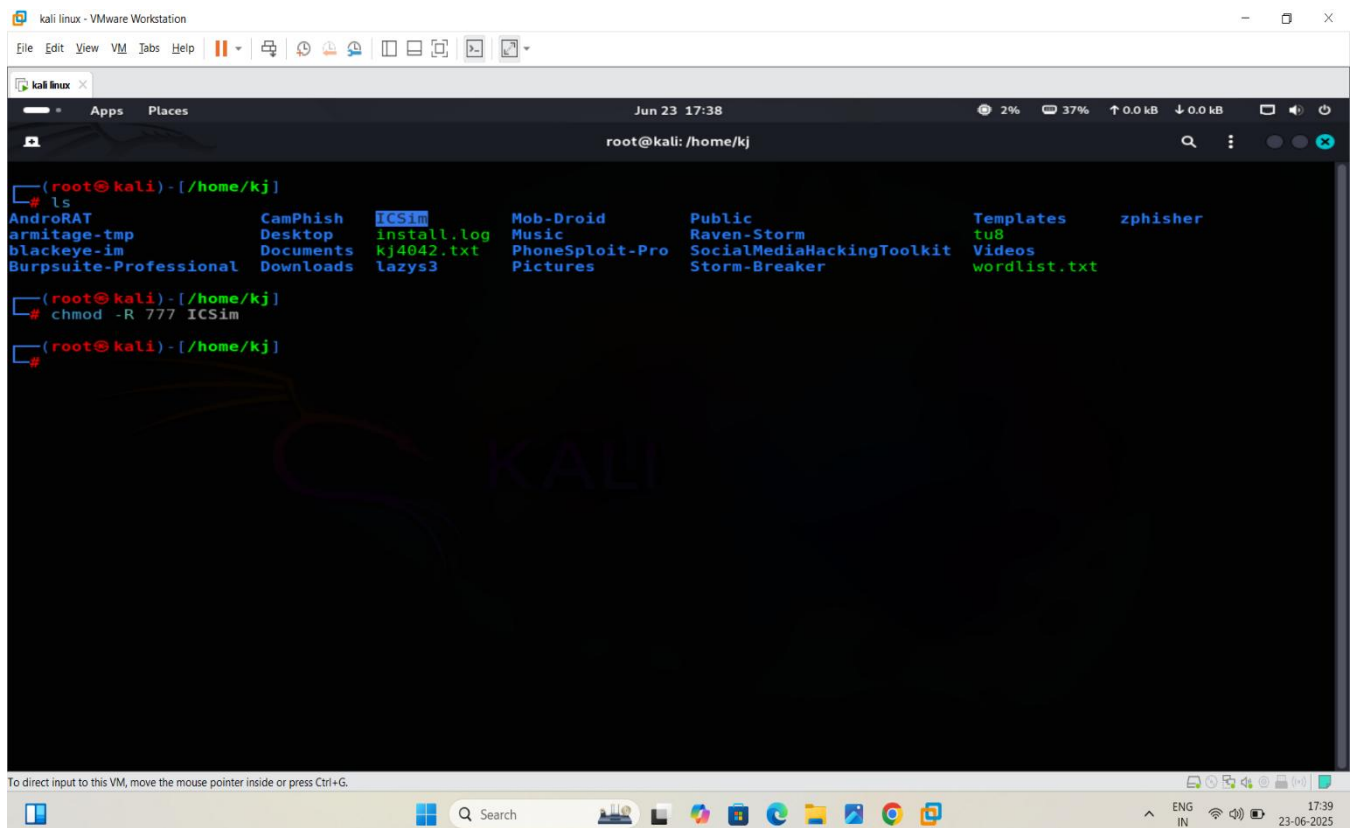
2nd step : open terminal and paste which you copy in github and paste command :

git clone <github copy section>

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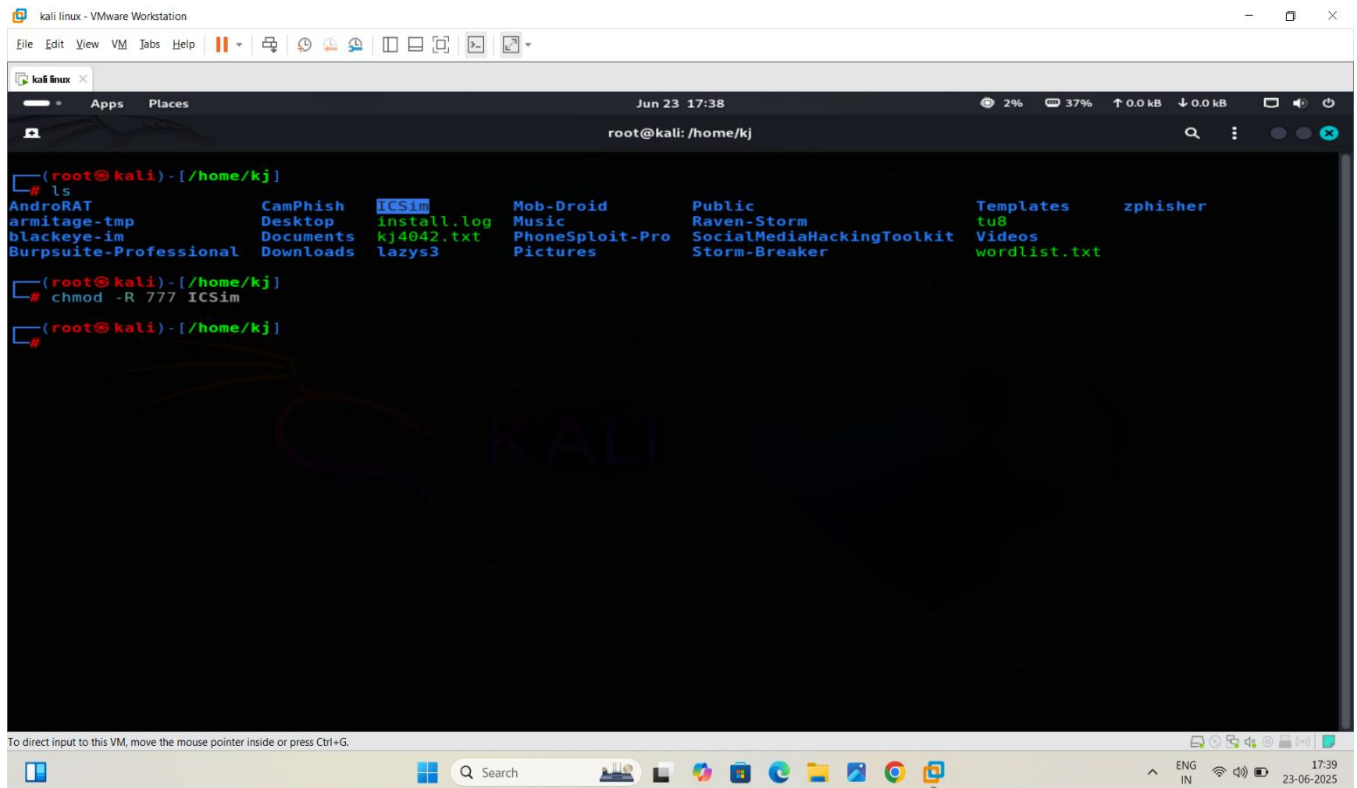


3. # ls : to check interface



4 . # chmod -R 777 ICSim

To give permission



The screenshot shows a Kali Linux terminal window within a VMware Workstation. The terminal prompt is `root@kali: ~/home/kj`. The user has run `ls`, displaying a directory listing of various tools and files. Then, the user runs `chmod -R 777 ICSim` to recursively set permissions on the `ICSim` directory. The terminal output shows the command being executed successfully.

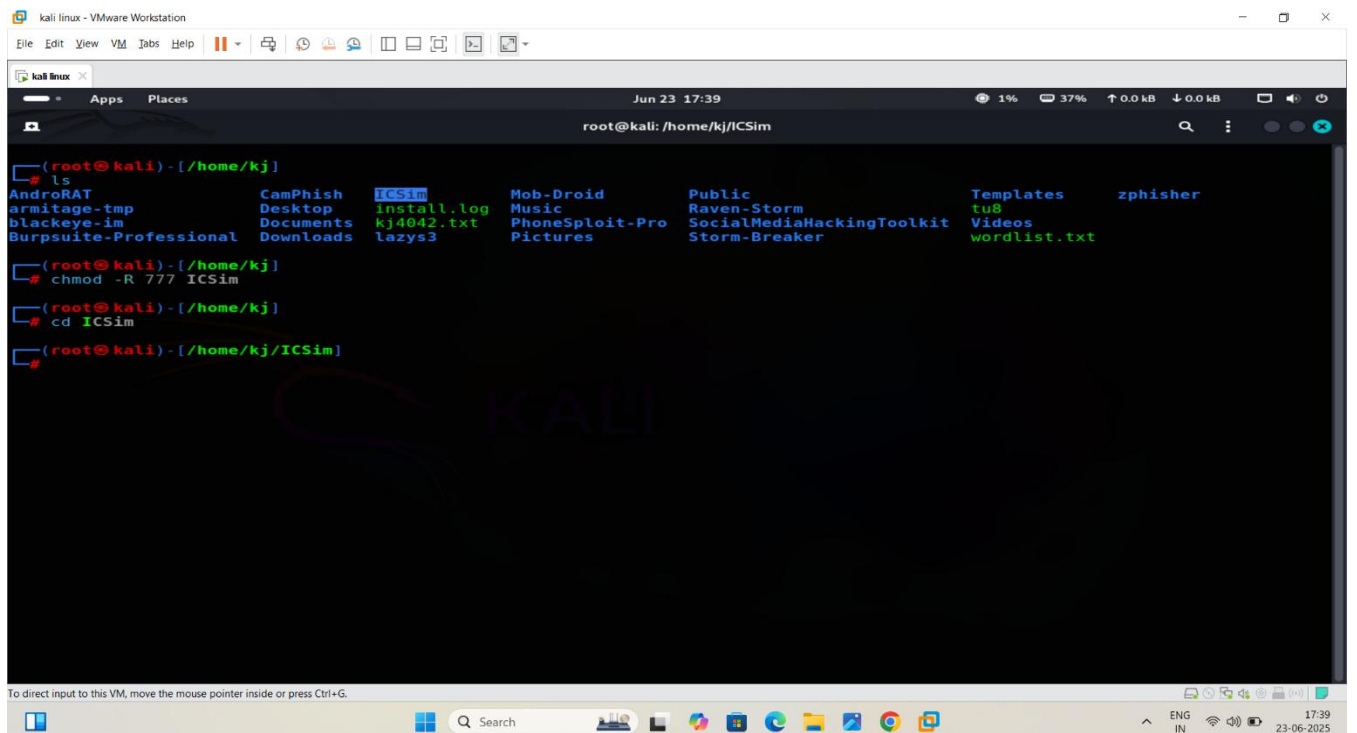
```
(root@kali) - [~/home/kj]
# ls
AndroRAT      CamPhish      ICSim          Mob-Droid      Public          Templates      zphisher
armitage-tmp  Desktop       install.log    Music          Raven-Storm    tu8            Videos
blackeye-im   Documents     kj4042.txt     PhoneSploit-Pro SocialMediaHackingToolkit wordlist.txt
Burpsuite-Professional Downloads     lazys3         Pictures       Storm-Breaker

(root@kali) - [~/home/kj]
# chmod -R 777 ICSim

(root@kali) - [~/home/kj]
```

5 . # cd ICSim

To go in ICSim directory



The screenshot shows the same Kali Linux terminal window. The user has run `cd ICSim` to change the current directory to `ICSim`. The terminal prompt now shows `root@kali: ~/home/kj/ICSim`. The directory listing from the previous screenshot is still visible in the background.

```
(root@kali) - [~/home/kj]
# ls
AndroRAT      CamPhish      ICSim          Mob-Droid      Public          Templates      zphisher
armitage-tmp  Desktop       install.log    Music          Raven-Storm    tu8            Videos
blackeye-im   Documents     kj4042.txt     PhoneSploit-Pro SocialMediaHackingToolkit wordlist.txt
Burpsuite-Professional Downloads     lazys3         Pictures       Storm-Breaker

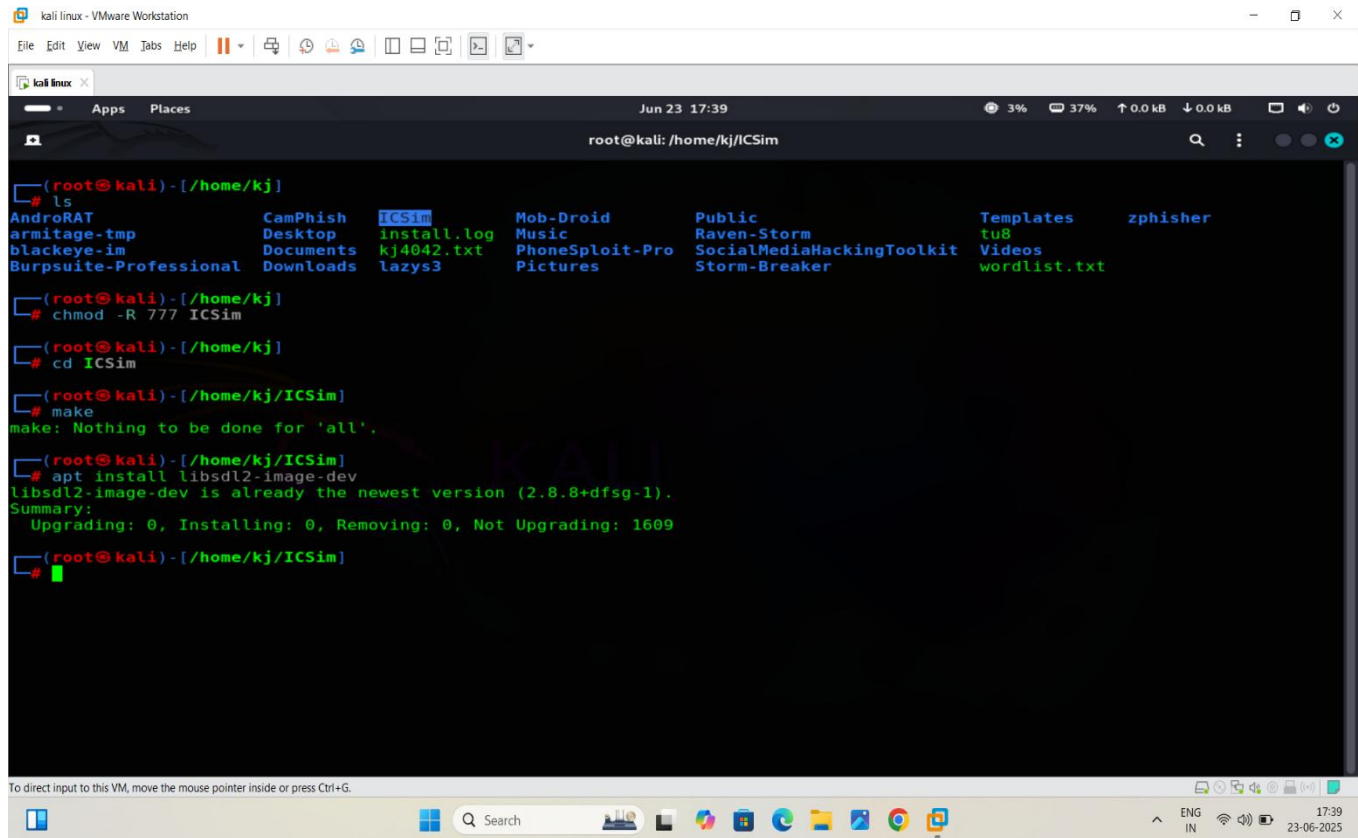
(root@kali) - [~/home/kj]
# chmod -R 777 ICSim

(root@kali) - [~/home/kj]
# cd ICSim

(root@kali) - [~/home/kj/ICSim]
#
```


6. # apt install libsd12-image-dev

libsd12-image-dev is required by ICSim to load and display PNG, JPG, and other image formats used in the simulation.



```
(root@kali) - [/home/kj]
# ls
AndroRAT          CamPhish  ICSim      Mob-Droid   Public      Templates    zphisher
armitage-tmp      Desktop   install.log Music        Raven-Storm SocialMediaHackingToolkit
blackeye-im       Documents kj4042.txt PhoneSploit-Pro
Burpsuite-Professional Downloads lazys3      Pictures    Storm-Breaker
                                                         tu8
                                                         Videos
                                                         wordlist.txt

(root@kali) - [/home/kj]
# chmod -R 777 ICSim

(root@kali) - [/home/kj]
# cd ICSim

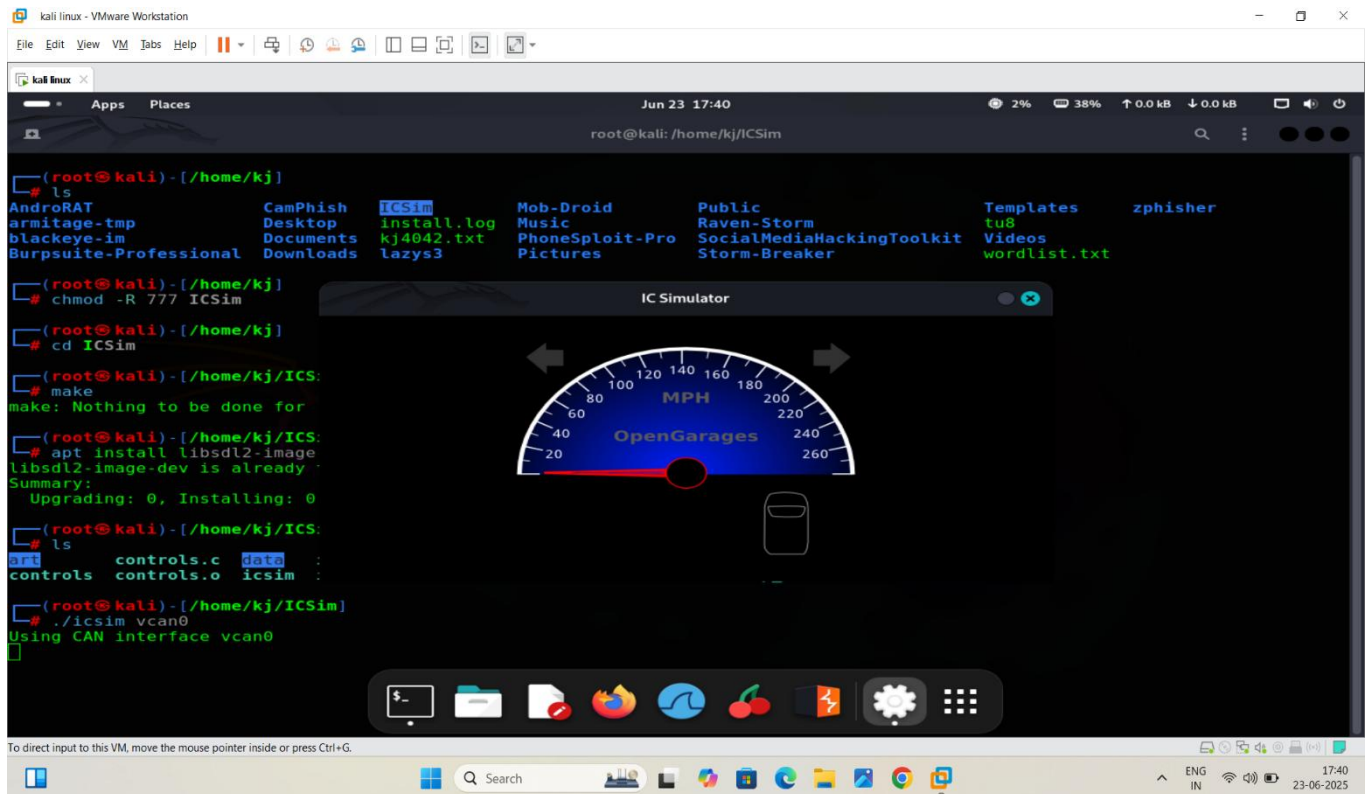
(root@kali) - [/home/kj/ICSim]
# make
make: Nothing to be done for 'all'.

(root@kali) - [/home/kj/ICSim]
# apt install libsd12-image-dev
libsd12-image-dev is already the newest version (2.8.8+dfsg-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1609

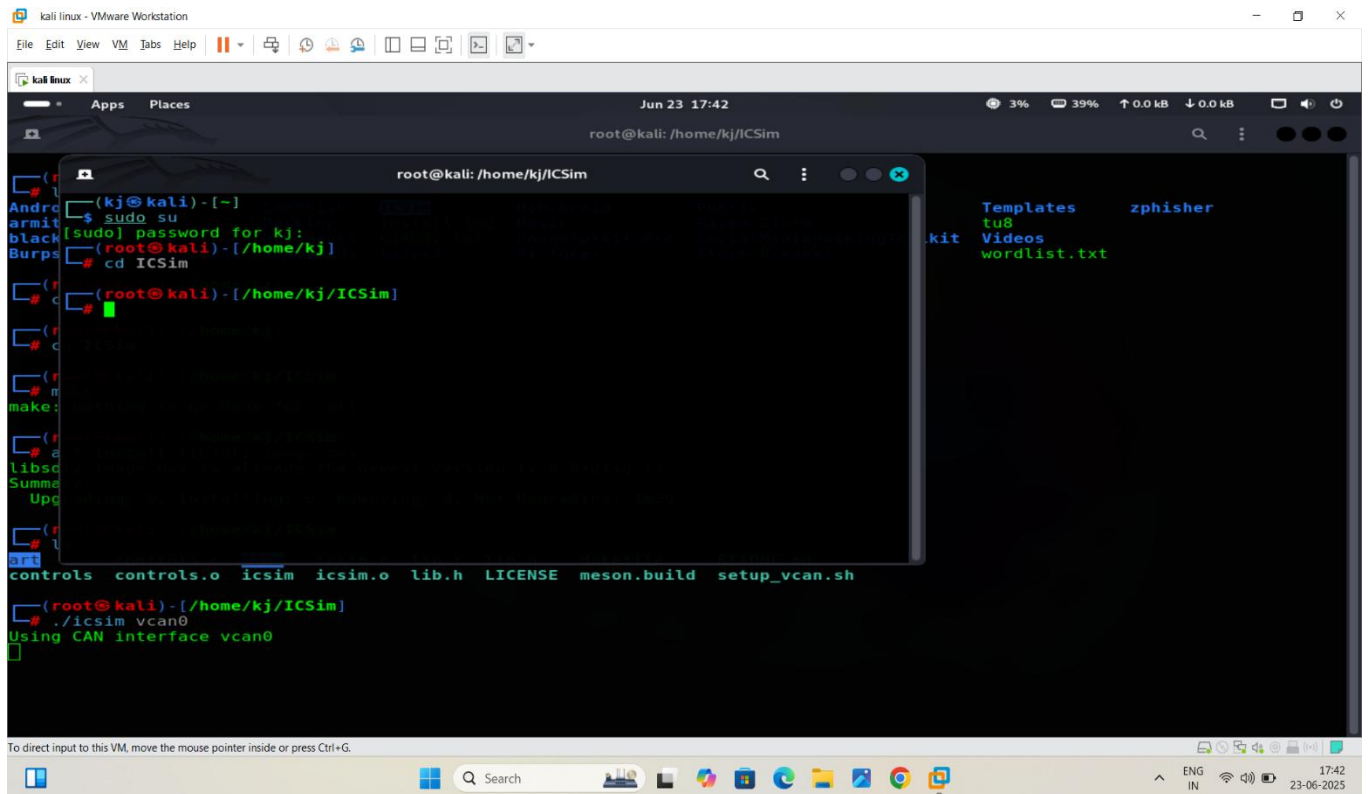
(root@kali) - [/home/kj/ICSim]
#
```

7. # ./icsim vcan0

You're starting the **Instrument Cluster Simulator (ICSim)** and telling it to connect to the **virtual CAN interface**

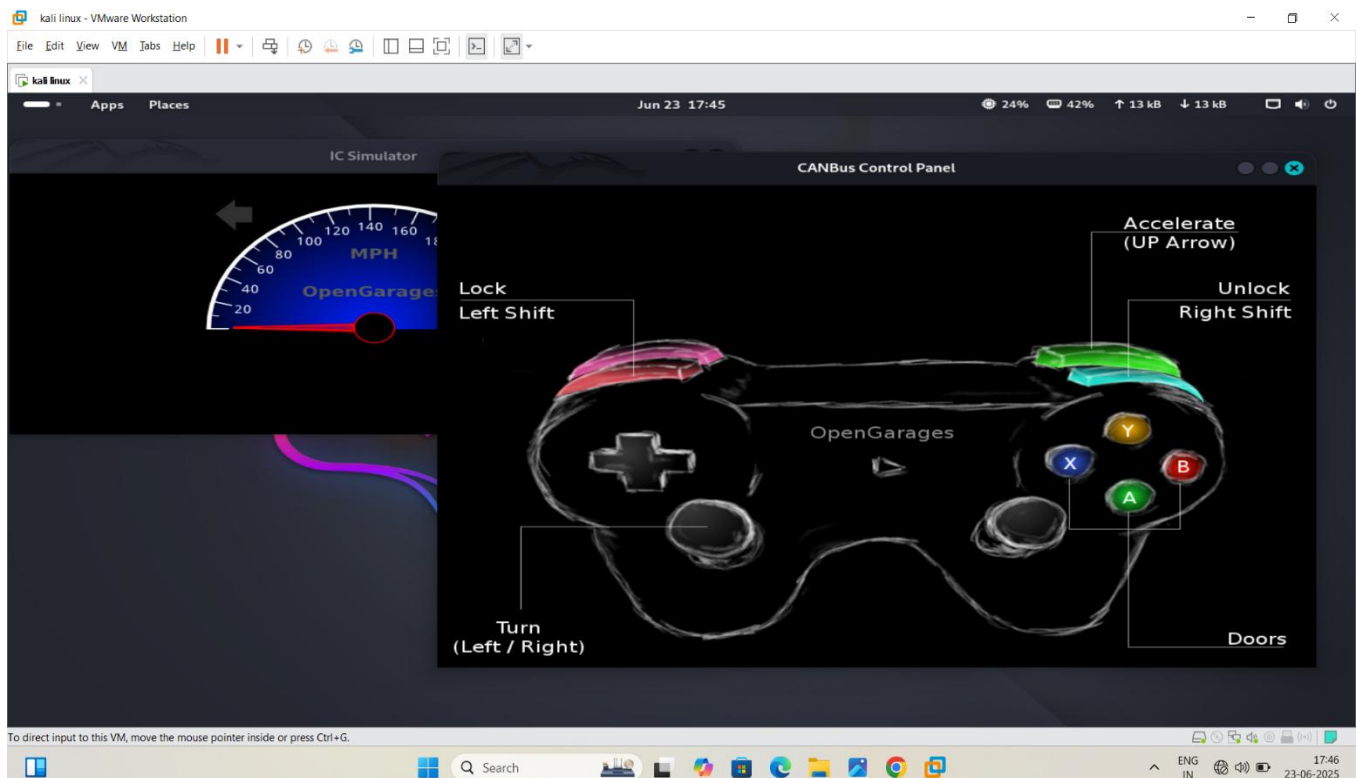
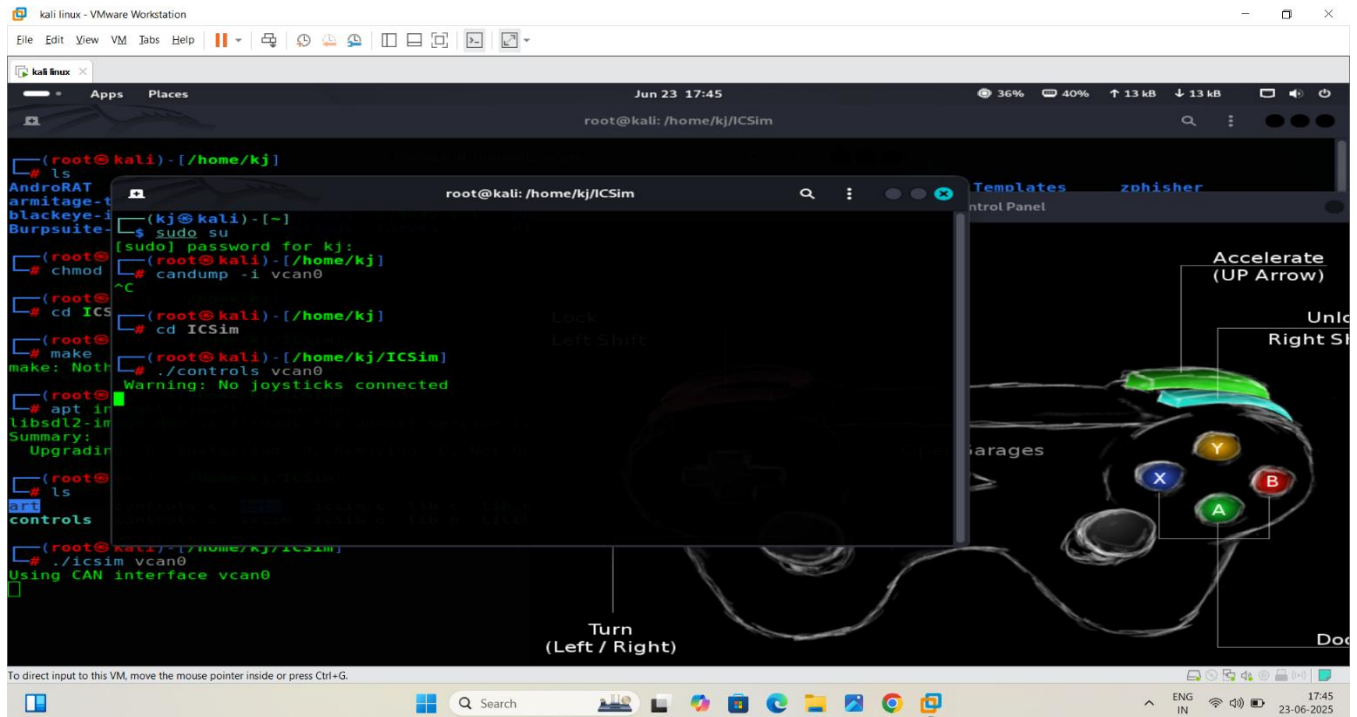


8. Open new terminal any go to ICSim directory using
cd ICSim

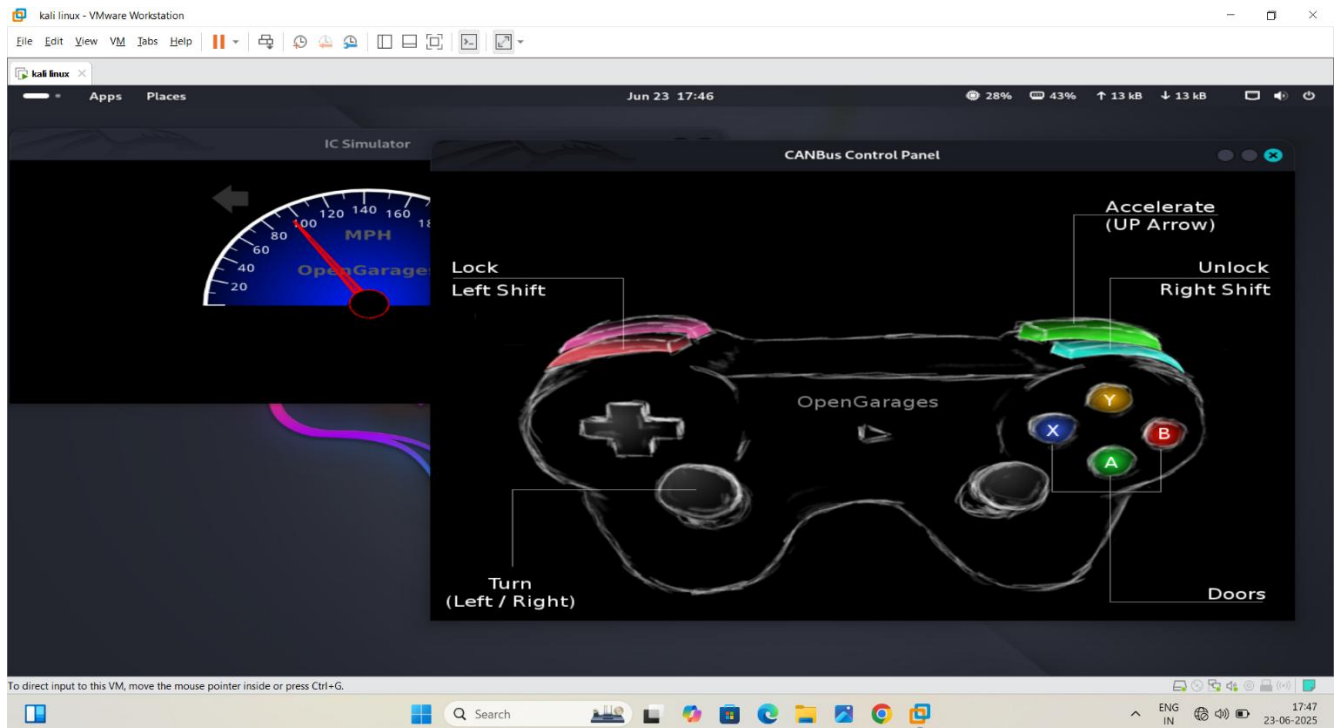


9. # ./controls vcan0

you're launching the **ICSim controls** interface, which allows you to send **simulated CAN messages** to the ICSim dashboard running on the same `vcan0` interface.

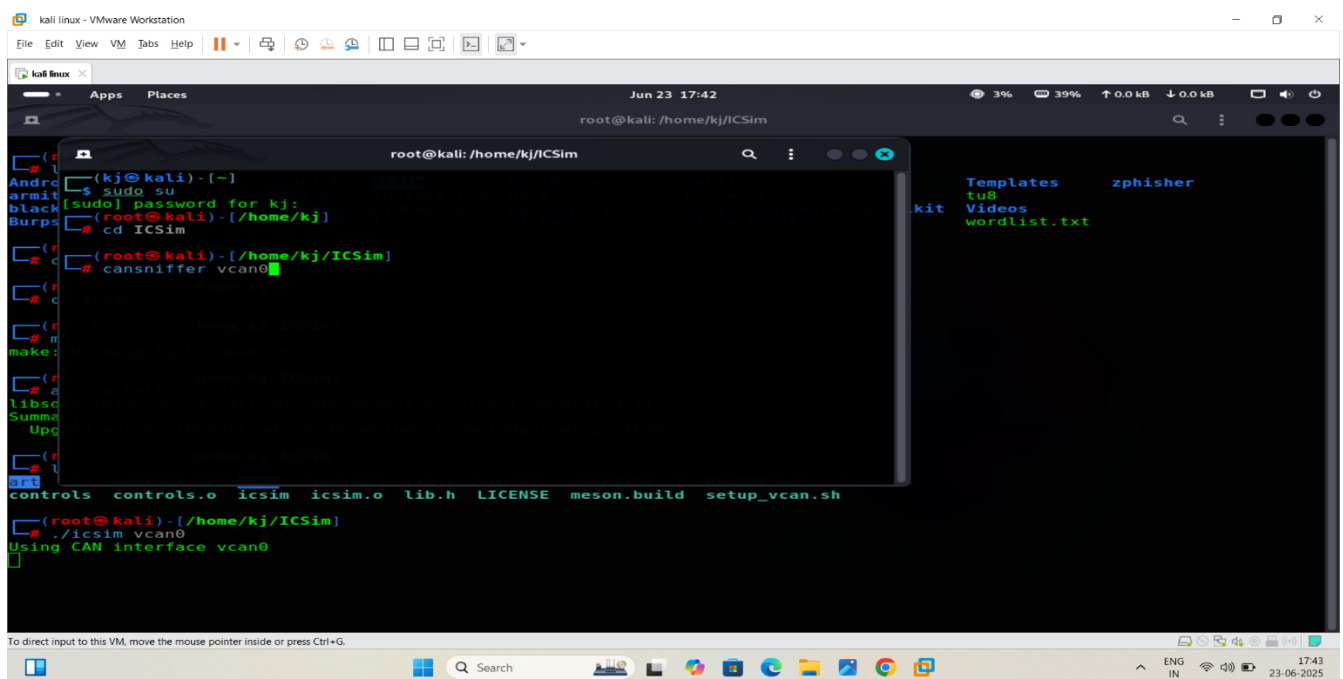


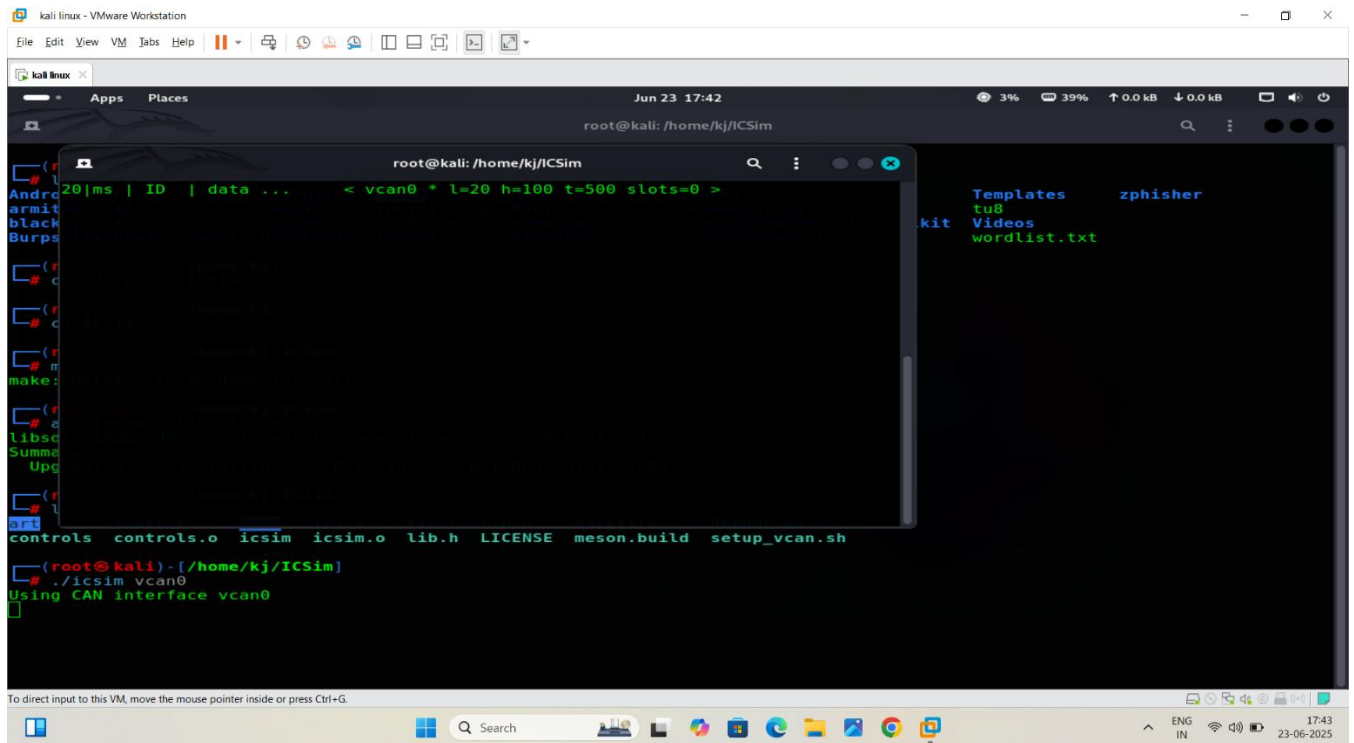
10. IF you click on up arrow of the keyboard that means u accelerate the car and check the speed meter is going high



11. open another terminal and come in ICSim directory

you're starting the **CAN bus sniffer tool** from `can-utils`, which monitors **live CAN traffic** on the `vcan0` interface and highlights **changing data** in real time.
`ansniffer vcan0`

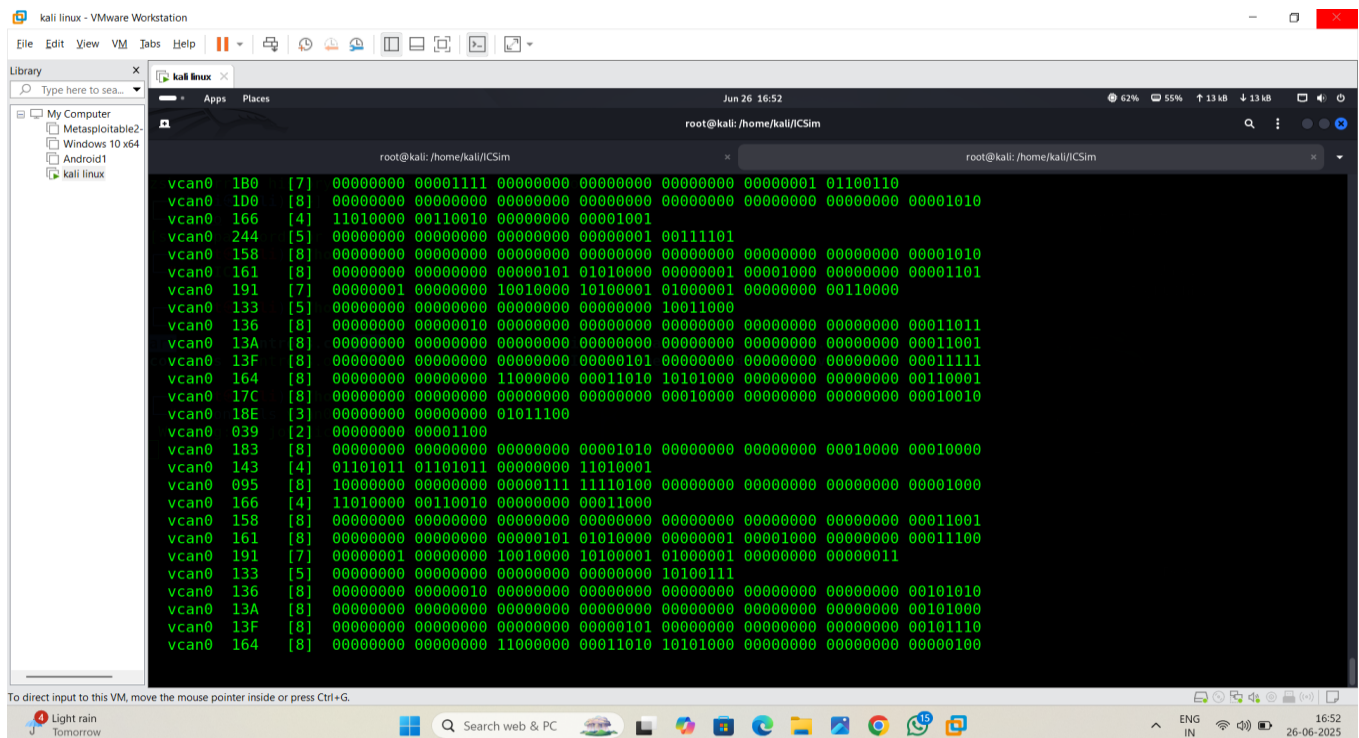




You can see the data changes here

12. # candump -l vcan0

This command listens to **CAN traffic on the vcan0 interface** and prints it live to the terminal.



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